INCH-POUND
MIL-DTL-32371/3
w/AMENDMENT 1
7 November 2023
SUPERSEDING
MIL-DTL-32371/3
24 April 2012

DETAIL SPECIFICATION SHEET

# FITTINGS, HYDRAULIC TUBE, SPHERICAL TO ORFS, MALE ADAPTER, 90° ELBOW

Inactive for new design after 13 March 2023.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-32371.

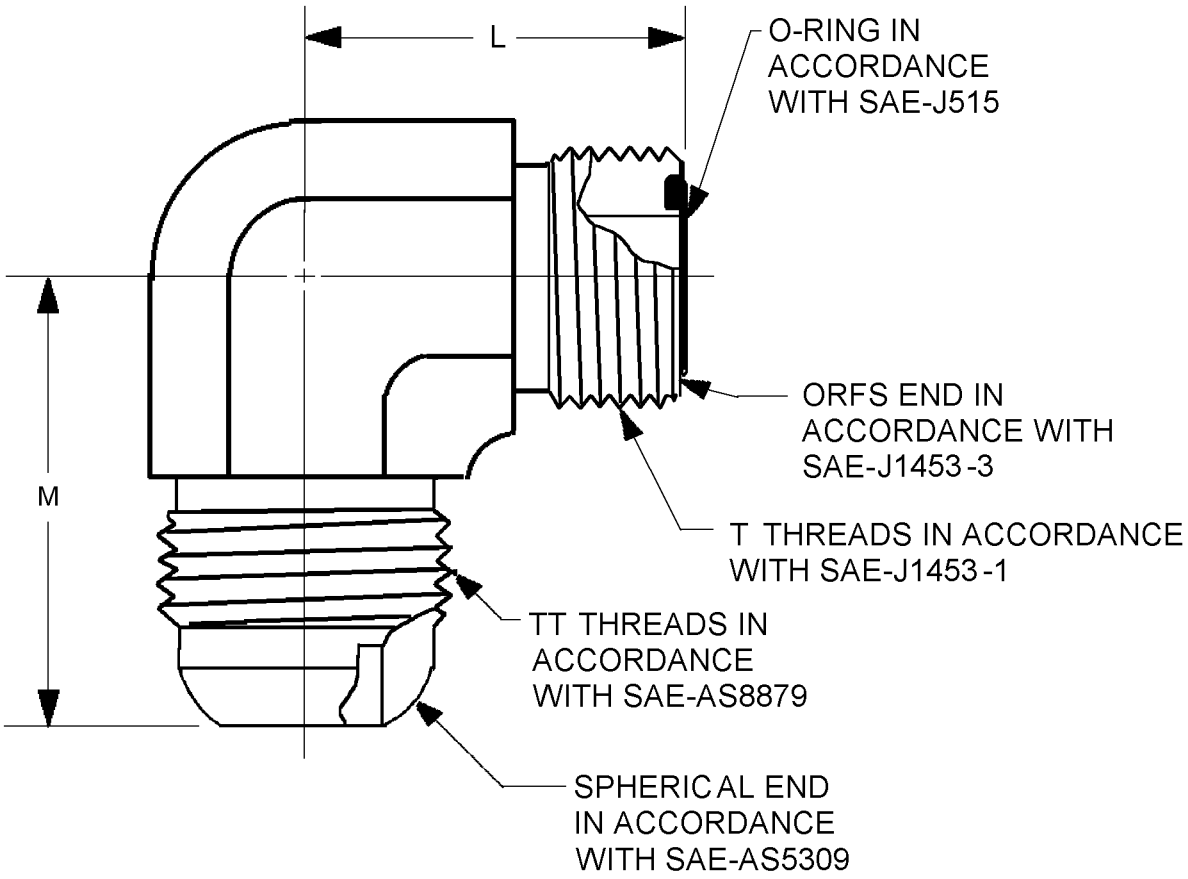



FIGURE 1. 90° elbow spherical to ORFS adapter.



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| Dash designator | Tube OD nom. inches (mm) | T ORFS thread | TT spherical Thread | L inches (mm) ±.04 (1.0) | M inches (mm) ±.04 (1.0) |
|---|---|---|---|---|---|
| -4 | .2500 (6.350) | 0.5625-18UNF-2A | .4375 - 20 UNJF-3A | .846 (21.50) | .970 (24.64) |
| -6 | .3750 (9.525) | 0.6875-16UN-2A | .5000-20 UNJF-3A | .984 (25.00) | 1.090 (27.69) |
| -8 | .5000 (12.700) | 0.8125-16UN-2A | .7500-16 UNJF-3A | 1.102 (28.00) | 1.360 (34.54) |
| -10 | .6250 (15.875) | 1.0000-14UN-2A | .8750-14 UNJF-3A | 1.319 (33.50) | 1.560 (39.62) |
| | | | | | |
| -12 | .7500 (19.050) | 1.1875-12UN-2A | 1.0625-12 UNJ-3A | 1.476 (37.50) | 1.780 (45.21) |
| -16 | 1.0000 (25.400) | 1.4375-12UN-2A | 1.3125-12 UNJ-3A | 1.634 (41.50) | 1.940 (49.28) |
| -20 | 1.2500 (31.750) | 1.6875-12UN-2A | 1.6250-12 UNJ-3A | 1.752 (44.50) | 2.170 (55.12) |
| -24 | 1.5000 (38.100) | 2.0000-12UN-2A | 1.8750-12 UNJ-3A | 1.929 (49.00) | 2.340 (59.44) |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Break all sharp edges and remove all burrs and slivers.
4. Dimensions and tolerances not shown shall be in accordance with SAE-AS5309 for spherical, 37° flared end, SAE-J1453-3 for ORFS end, and SAE-J514 for body.
5. This drawing is for identification purposes only and is not intended to restrict designs and shapes not dimensioned.

FIGURE 1. 90° elbow spherical to ORFS adapter - Continued.

REQUIREMENTS:

Fittings shall be as specified on figure 1 and in table I.

Materials and finishes shall be in accordance with MIL-DTL-32371.

Finish. Finishes shall be as specified in table I. All platings shall be capable of meeting a minimum of 96 hours salt spray test in accordance with ASTM B117. The fittings shall show no evidence of corrosion after 96 hours of salt spray. Fluid passages, other openings, and internal threads shall not be subject to the plating thickness requirement and may have bare areas provided they are protected with a light film of oil.



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TABLE I. Material and chemical finish identification codes and chemical finish reference specifications. 1/

| PIN code material/plating finish | Material | Plating finish |
|---|---|---|
| Blank | Steel | Cadmium plating in accordance with SAE-AMS-C-81562, type II, class 3 or SAE-AMS-QQ-P-416, type II, class 2. 2/ |
| CN | Steel | Cadmium plating in accordance with SAE-AMS-C-81562, type II, class 3 or SAE-AMS-QQ-P-416, type II, class 2 with NAVAIR trivalent chromium pretreatment (TCP) in accordance with MIL-DTL-81706, type II, class 1A. |
| E | Steel | NAVAIR TCP in accordance with MIL-DTL-81706, type II, class 1A. |
| F | Steel | Zinc plate (finish J, P, or R) with NAVAIR TCP in accordance with MIL-DTL-81706, type II, class 1A. |
| H | Steel | Aluminum-nickel in accordance with ASTM F1136/F1136M, grade 3, NC. |
| J | Steel | Zinc-nickel in accordance with SAE-AMS2417, type 2, grade B. |
| M | Nickel-copper alloy UNS N04400 | No additional finish. |
| N | High-chromium nickel alloy UNS N06690 | No additional finish. |
| P | Steel | Zinc phosphate finish in accordance MIL-DTL-16232 type Z, class1. |
| R | Steel | Zinc plating in accordance with ASTM B633; type VI, Fe/Zn 5. 3/ |
| S | Corrosion resistant steel | No additional finish. Passivation in accordance with SAE-AMS2700, type 6 or 7. |
| T | Titanium 4/ | Anodize in accordance with SAE-AMS2488 type. |
| TF | Titanium 4/ | Fluoride phosphate in accordance with SAE-AMS2486. |
| Z | Steel | Zinc plating in accordance with ASTM B633; type II or III, Fe/Zn 5, or ASTM B695, type II, class 5. |
| ZN | Steel | Zinc plating in accordance with ASTM B633; type II or III, Fe/Zn 5, or ASTM B695, type II, class 5 with NAVAIR TCP in accordance with MIL-DTL-81706, type II, class 1A. |

1/ Table is for Part or Identifying Number (PIN) code use only. Materials and finishes shall be in accordance with MIL-DTL-32371.
2/ Hydrogen embrittlement relief test need not be run.
3/ Hexavalent chromium free.
4/ Titanium shall not be used in oxygen or potable water systems.

Trivalent wrenchability. When the finish has been damaged due to poor wrenchability, the surface of the connector shall be touched up using the brush plating process below. The term “trivalent wrenchability” is used to evaluate the ability of the finish to withstand abrasion from an excessive amount of wrenching.

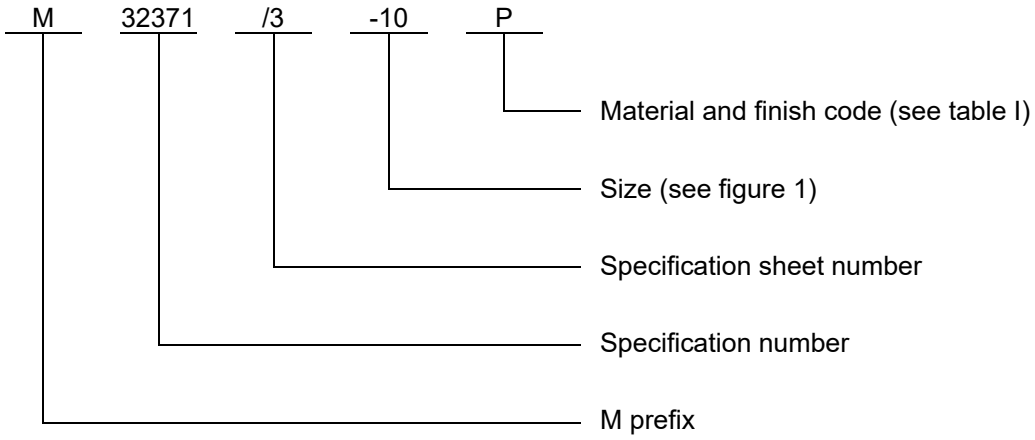
a. Brush plating of hard chromium by electrodeposition shall be in accordance with SAE-AMS2451/5.
b. Brush plating of medium-hardness, low stress nickel by electrodeposition shall be in accordance with SAE-AMS2451/9.
c. Brush plating of NAVAIR TCP shall be in accordance with MIL-DTL-81706, type II, class 1A, material form 1 through 6, application method B. Example of a PIN: M817062A6B.

Maximum operating pressure. Maximum operating pressure shall be in accordance with MIL-DTL-32371.



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PIN: The PIN consists of the letter “M”, the specification number, a dash number, and a material and finish code letter.




PIN example: M32371/3-10P indicates a 90° male 90° elbow adapter, .6250 inch (.1588 mm), steel with NAVAIR TCP.

O-ring is not supplied order separately, see SAE-J515.

Cadmium is not recommended. To the users of this document, it is recommended that the use of carbon steel material with cadmium plating be used only when other materials and finishes specified in this document cannot meet performance requirements.

Referenced documents. In addition to MIL-DTL-32371, this document references the following:

| | | |
|---|---|---|
| MIL-DTL-16232 | SAE-AMS-QQ-P-416 | SAE-AS8879 |
| MIL-DTL-81706 | SAE-AMS2417 | SAE-AS5309 |
| ASTM B117 | SAE-AMS2451/5 | SAE-J514 |
| ASTM B633 | SAE-AMS2451/9 | SAE-J515 |
| ASTM B695 | SAE-AMS2486 | SAE-J1453-1 |
| ASTM F1136/F1136M | SAE-AMS2488 | SAE-J1453-3 |
| SAE-AMS-C-81562 | SAE-AMS2700 | |

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.



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CONCLUDING MATERIAL

Custodians:
Army - AT
Navy - AS
Air Force - 71
DLA - CC

Preparing activity:
DLA - CC

(Project 4730-2023-055)

Review activities:
Army - AR
Navy - CG, MC, SA, SH

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